DE — resultados CEC

Benchmark MAREA · paper OE1 · 71 funciones CEC2014/2017/2022 · 51 corridas por funcion

Protocolo

Valores = **error al optimo**, f(x) - f*. Menor es mejor; 0 es el optimo global. Dominio [-100,100] en todas las funciones. Poblacion N=80, iteraciones T=MaxFES/N. Presupuestos 5.000 / 50.000 / 500.000 evaluaciones. Las 30 funciones de CEC2014 y las 29 de CEC2017 corren en D=10 y D=50; las 12 de CEC2022 solo llegan a D=20, asi que en la columna D50 corren en **D=20** (marcado en la tabla). Todos los resultados vienen del mismo entorno del Mac mini (python 3.11.9, numpy 2.3.3, opfunu 1.0.4).

Resumen — posicion entre los 11 algoritmos

dimension	presupuesto	rango medio de Friedman	posicion de 11	funciones donde es el mejor
D10	5k FES	6.13	6º	4 de 71
D10	50k FES	3.03	2º	27 de 71
D10	500k FES	2.12	1º	41 de 71
D50	5k FES	8.11	9º	0 de 71
D50	50k FES	7.45	8º	3 de 71
D50	500k FES	5.82	6º	11 de 71

DE — estadisticas por funcion, D = 10

error al optimo sobre 51 corridas · b=best w=worst med=median avg=mean sd=std

func	suite	dim	5k					50k					500k				
			b	w	med	avg	sd	b	w	med	avg	sd	b	w	med	avg	sd
F12014	2014	10	1.22e+06	1.09e+07	5.84e+06	5.99e+06	2.30e+06	7.58e+02	5.15e+03	2.17e+03	2.33e+03	1.04e+03	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F22014	2014	10	3.02e+08	1.39e+09	6.77e+08	7.36e+08	2.28e+08	3.91e+02	4.65e+03	1.14e+03	1.33e+03	7.50e+02	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F32014	2014	10	2.04e+03	1.05e+04	5.68e+03	5.50e+03	1.49e+03	6.14e-03	4.99e-02	2.59e-02	2.62e-02	1.17e-02	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F42014	2014	10	4.34e+01	1.07e+02	7.66e+01	7.82e+01	1.71e+01	1.36e-02	3.48e+01	4.34e+00	1.25e+01	1.60e+01	0.00e+00	3.48e+01	4.34e+00	1.24e+01	1.60e+01
F52014	2014	10	2.02e+01	2.08e+01	2.05e+01	2.05e+01	1.21e-01	2.01e+01	2.04e+01	2.02e+01	2.02e+01	9.25e-02	2.01e+01	2.03e+01	2.02e+01	2.02e+01	4.93e-02
F62014	2014	10	7.27e+00	1.17e+01	9.45e+00	9.46e+00	8.75e-01	9.99e-01	6.11e+00	2.87e+00	3.00e+00	1.27e+00	0.00e+00	4.54e+00	8.95e-01	1.09e+00	1.07e+00
F72014	2014	10	3.89e+00	1.20e+01	6.93e+00	7.32e+00	1.84e+00	3.37e-01	7.53e-01	6.05e-01	5.96e-01	9.47e-02	9.86e-03	4.72e-01	1.16e-01	1.36e-01	1.00e-01
F82014	2014	10	3.25e+01	6.67e+01	5.44e+01	5.40e+01	6.83e+00	1.86e+01	3.89e+01	2.92e+01	2.90e+01	4.61e+00	0.00e+00	1.15e+01	9.95e-01	1.15e+00	1.95e+00
F92014	2014	10	4.65e+01	8.08e+01	6.36e+01	6.37e+01	6.83e+00	2.09e+01	5.01e+01	4.04e+01	3.98e+01	5.84e+00	1.27e+00	2.78e+01	7.96e+00	9.19e+00	5.72e+00
F102014	2014	10	6.51e+02	1.31e+03	1.04e+03	1.03e+03	1.34e+02	1.21e+02	5.52e+02	2.88e+02	2.99e+02	1.06e+02	2.16e+01	7.04e+01	4.23e+01	4.40e+01	1.29e+01
F112014	2014	10	1.48e+03	2.06e+03	1.81e+03	1.78e+03	1.57e+02	1.15e+02	1.52e+03	7.89e+02	8.51e+02	3.86e+02	6.16e+01	6.60e+02	1.16e+02	1.89e+02	1.50e+02
F122014	2014	10	1.05e+00	2.32e+00	1.62e+00	1.63e+00	2.95e-01	8.60e-02	8.80e-01	3.41e-01	3.87e-01	2.25e-01	5.85e-02	4.33e-01	1.70e-01	1.88e-01	7.92e-02
F132014	2014	10	4.34e-01	9.94e-01	7.62e-01	7.44e-01	1.25e-01	2.02e-01	4.07e-01	3.21e-01	3.14e-01	5.39e-02	2.92e-02	2.54e-01	1.57e-01	1.39e-01	6.41e-02
F142014	2014	10	4.52e-01	2.85e+00	1.16e+00	1.29e+00	5.77e-01	1.28e-01	2.90e-01	2.23e-01	2.22e-01	4.37e-02	3.15e-02	2.15e-01	1.24e-01	1.20e-01	4.31e-02
F152014	2014	10	1.44e+01	3.61e+02	8.79e+01	1.00e+02	7.61e+01	1.71e+00	4.57e+00	3.43e+00	3.40e+00	6.43e-01	3.26e-01	2.39e+00	1.19e+00	1.30e+00	5.45e-01
F162014	2014	10	3.56e+00	4.21e+00	3.98e+00	3.97e+00	1.37e-01	2.81e+00	3.81e+00	3.49e+00	3.48e+00	2.05e-01	2.34e-01	2.75e+00	1.39e+00	1.53e+00	7.58e-01
F172014	2014	10	1.10e+04	5.67e+04	2.64e+04	2.80e+04	1.06e+04	7.80e+01	4.23e+02	2.00e+02	2.12e+02	8.35e+01	0.00e+00	2.20e+00	2.08e-01	2.12e-01	3.88e-01
F182014	2014	10	3.17e+03	1.15e+04	7.65e+03	7.67e+03	2.07e+03	6.29e+00	2.25e+01	1.33e+01	1.38e+01	3.13e+00	1.18e-03	1.01e+00	9.93e-03	3.24e-02	1.40e-01
F192014	2014	10	1.04e+01	1.47e+02	3.21e+01	3.79e+01	2.55e+01	2.65e+00	4.70e+00	3.68e+00	3.68e+00	4.22e-01	3.66e-02	2.42e+00	1.70e+00	1.60e+00	5.36e-01
F202014	2014	10	1.11e+04	3.72e+05	1.14e+05	1.33e+05	8.78e+04	1.01e+01	5.88e+01	2.56e+01	2.73e+01	8.75e+00	2.71e-05	4.72e-01	3.20e-03	4.28e-02	1.03e-01
F212014	2014	10	4.76e+03	4.60e+04	2.12e+04	2.12e+04	9.52e+03	6.12e+00	1.18e+02	4.30e+01	4.50e+01	2.57e+01	2.78e-05	3.31e-01	7.36e-03	7.29e-02	1.30e-01
F222014	2014	10	3.95e+01	3.45e+02	2.00e+02	2.09e+02	6.56e+01	2.50e+01	3.26e+01	2.87e+01	2.89e+01	1.71e+00	7.84e-04	2.10e+01	6.16e-01	3.73e+00	6.95e+00
F232014	2014	10	2.59e+02	2.83e+02	2.70e+02	2.70e+02	5.96e+00	2.37e+02	2.37e+02	2.37e+02	2.37e+02	4.01e-05	2.37e+02	2.37e+02	2.37e+02	2.37e+02	5.74e-14
F242014	2014	10	1.48e+02	1.90e+02	1.74e+02	1.73e+02	9.02e+00	1.29e+02	1.55e+02	1.48e+02	1.47e+02	5.52e+00	1.00e+02	1.26e+02	1.13e+02	1.13e+02	5.39e+00
F252014	2014	10	2.00e+02	2.03e+02	2.02e+02	2.02e+02	5.21e-01	1.50e+02	2.00e+02	2.00e+02	1.99e+02	7.08e+00	1.40e+02	2.00e+02	2.00e+02	1.99e+02	8.45e+00
F262014	2014	10	1.00e+02	1.01e+02	1.01e+02	1.01e+02	1.47e-01	1.00e+02	1.00e+02	1.00e+02	1.00e+02	5.15e-02	1.00e+02	1.00e+02	1.00e+02	1.00e+02	7.30e-02
F272014	2014	10	1.52e+02	5.60e+02	4.25e+02	4.18e+02	8.97e+01	5.47e+00	4.00e+02	4.00e+02	3.41e+02	1.36e+02	2.13e+00	4.00e+02	4.00e+02	3.40e+02	1.38e+02
F282014	2014	10	3.74e+02	4.11e+02	3.86e+02	3.86e+02	6.44e+00	3.69e+02	4.04e+02	3.78e+02	3.78e+02	8.48e+00	3.69e+02	4.04e+02	3.78e+02	3.78e+02	8.48e+00
F292014	2014	10	4.65e+06	6.57e+06	5.37e+06	5.34e+06	4.32e+05	4.50e+06	5.31e+06	4.50e+06	4.63e+06	2.98e+05	4.50e+06	5.31e+06	4.50e+06	4.63e+06	2.98e+05
F302014	2014	10	3.46e+04	2.74e+06	4.13e+05	6.11e+05	5.53e+05	2.44e+03	4.90e+04	3.32e+03	4.65e+03	6.51e+03	1.72e+03	4.90e+04	2.31e+03	3.18e+03	6.55e+03
F12017	2017	10	2.28e+08	9.39e+08	5.30e+08	5.60e+08	1.60e+08	7.57e+02	8.40e+03	2.85e+03	2.93e+03	1.50e+03	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F22017	2017	10	1.62e+03	9.64e+03	5.68e+03	5.84e+03	1.62e+03	2.43e-03	3.38e-02	1.19e-02	1.23e-02	6.82e-03	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F32017	2017	10	3.21e+01	1.68e+02	1.04e+02	1.04e+02	2.84e+01	2.72e-02	4.13e+00	1.24e-01	6.70e-01	1.37e+00	0.00e+00	3.99e+00	0.00e+00	5.47e-01	1.39e+00
F42017	2017	10	4.96e+02	1.52e+03	9.41e+02	9.80e+02	2.32e+02	2.46e+01	6.15e+01	4.49e+01	4.44e+01	6.71e+00	9.95e-01	2.91e+01	6.93e+00	9.42e+00	6.88e+00
F52017	2017	10	2.45e-04	6.17e-03	2.10e-03	2.37e-03	1.31e-03	3.13e-06	7.35e-04	3.77e-04	3.68e-04	1.70e-04	0.00e+00	1.47e-04	3.03e-07	6.37e-06	2.13e-05
F62017	2017	10	7.06e+03	3.09e+04	1.83e+04	1.76e+04	5.09e+03	4.57e+01	6.82e+01	5.85e+01	5.88e+01	5.27e+00	1.99e+00	4.23e+01	1.85e+01	2.05e+01	7.24e+00
F72017	2017	10	5.84e+01	1.40e+02	1.04e+02	1.02e+02	1.89e+01	3.11e+01	6.71e+01	5.59e+01	5.42e+01	8.01e+00	1.00e+01	4.80e+01	2.01e+01	2.08e+01	7.39e+00
F82017	2017	10	1.20e+00	3.51e+00	2.35e+00	2.36e+00	6.31e-01	1.07e-06	7.54e-05	8.17e-06	1.29e-05	1.47e-05	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F92017	2017	10	1.39e+03	2.06e+03	1.78e+03	1.74e+03	1.61e+02	4.88e+02	1.52e+03	1.10e+03	1.04e+03	2.80e+02	3.29e+01	5.24e+02	1.30e+02	1.38e+02	1.01e+02
F102017	2017	10	1.82e+04	9.56e+04	3.63e+04	3.76e+04	1.50e+04	1.15e+01	9.48e+01	2.11e+01	2.37e+01	1.23e+01	0.00e+00	1.99e+00	0.00e+00	2.15e-01	4.59e-01
F112017	2017	10	2.94e+06	4.49e+07	2.06e+07	2.07e+07	9.72e+06	1.32e+03	6.24e+03	2.62e+03	2.65e+03	9.98e+02	0.00e+00	2.08e-01	0.00e+00	5.71e-02	9.38e-02
F122017	2017	10	4.93e+04	8.51e+05	2.72e+05	2.91e+05	1.91e+05	1.65e+01	6.62e+01	3.09e+01	3.20e+01	8.59e+00	0.00e+00	8.84e+00	4.66e+00	4.38e+00	2.36e+00
F132017	2017	10	5.00e+02	3.74e+03	2.25e+03	2.31e+03	7.26e+02	3.10e+01	6.00e+01	4.56e+01	4.56e+01	6.27e+00	5.04e+00	2.90e+01	2.20e+01	1.99e+01	6.29e+00
F142017	2017	10	7.13e+03	5.39e+04	3.22e+04	3.21e+04	1.03e+04	1.31e+01	6.41e+01	2.57e+01	2.94e+01	1.17e+01	4.43e-08	1.00e+00	1.55e-03	2.25e-02	1.40e-01
F152017	2017	10	8.06e+02	3.48e+03	1.74e+03	1.88e+03	6.85e+02	2.83e-01	5.29e+00	1.19e+00	1.21e+00	6.34e-01	4.54e-08	2.39e-01	2.28e-01	1.29e-01	1.07e-01
F162017	2017	10	2.70e+02	1.57e+03	7.56e+02	7.93e+02	2.93e+02	3.32e+01	5.50e+01	4.21e+01	4.23e+01	4.77e+00	3.12e-01	2.75e+01	2.00e+01	1.56e+01	8.12e+00
F172017	2017	10	1.02e+04	4.83e+04	2.46e+04	2.54e+04	9.67e+03	1.38e+01	3.00e+01	2.60e+01	2.51e+01	3.45e+00	7.16e-08	8.01e-03	8.75e-04	1.95e-03	2.25e-03
F182017	2017	10	6.05e+02	8.22e+03	3.09e+03	3.66e+03	2.16e+03	1.56e+00	4.00e+00	2.43e+00	2.47e+00	5.52e-01	4.97e-02	1.02e+00	3.79e-01	4.33e-01	2.50e-01
F192017	2017	10	6.69e+01	4.75e+02	1.88e+02	2.05e+02	8.81e+01	2.48e+01	4.25e+01	3.46e+01	3.37e+01	4.65e+00	7.95e-04	1.42e+01	1.11e-03	5.05e+00	5.88e+00

func	suite	dim	5k					50k					500k				
			b	w	med	avg	sd	b	w	med	avg	sd	b	w	med	avg	sd
F202017	2017	10	3.80e+02	5.86e+02	4.90e+02	4.89e+02	3.95e+01	1.00e+02	1.00e+02	1.00e+02	1.00e+02	2.84e-02	1.00e+02	1.00e+02	1.00e+02	1.00e+02	0.00e+00
F212017	2017	10	1.14e+02	1.19e+02	1.17e+02	1.17e+02	1.09e+00	1.00e+02	1.06e+02	1.04e+02	1.04e+02	1.13e+00	1.00e+02	1.00e+02	1.00e+02	1.00e+02	0.00e+00
F222017	2017	10	5.59e+02	9.37e+02	7.94e+02	7.65e+02	9.07e+01	1.00e+02	1.01e+02	1.00e+02	1.00e+02	1.14e-01	1.00e+02	1.00e+02	1.00e+02	1.00e+02	0.00e+00
F232017	2017	10	9.52e+02	1.64e+03	1.26e+03	1.28e+03	1.29e+02	2.00e+02	2.00e+02	2.00e+02	2.00e+02	6.66e-02	2.00e+02	2.00e+02	2.00e+02	2.00e+02	0.00e+00
F242017	2017	10	5.40e+02	6.49e+02	5.76e+02	5.82e+02	2.78e+01	4.80e+02	4.81e+02	4.80e+02	4.80e+02	1.66e-01	4.76e+02	4.80e+02	4.79e+02	4.79e+02	4.60e-01
F252017	2017	10	4.58e+02	4.73e+02	4.66e+02	4.65e+02	4.20e+00	4.55e+02	4.67e+02	4.56e+02	4.59e+02	5.14e+00	4.55e+02	4.67e+02	4.56e+02	4.59e+02	5.14e+00
F262017	2017	10	3.95e+02	4.06e+02	4.02e+02	4.01e+02	2.34e+00	3.90e+02	3.91e+02	3.90e+02	3.90e+02	4.90e-01	3.90e+02	3.91e+02	3.90e+02	3.90e+02	4.90e-01
F272017	2017	10	1.93e+02	3.27e+02	2.47e+02	2.49e+02	2.62e+01	1.03e-01	4.21e-01	2.11e-01	2.21e-01	7.43e-02	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F282017	2017	10	2.31e+03	3.74e+04	6.64e+03	7.80e+03	5.91e+03	9.98e+02	1.03e+03	1.01e+03	1.02e+03	8.01e+00	9.79e+02	9.83e+02	9.79e+02	9.79e+02	9.50e-01
F292017	2017	10	1.52e+05	3.84e+07	3.73e+06	5.58e+06	5.89e+06	2.33e+03	1.00e+04	5.39e+03	5.54e+03	1.67e+03	1.38e+03	1.38e+03	1.38e+03	1.38e+03	1.69e-01
F12022	2022	10	2.27e+03	7.52e+03	4.59e+03	4.67e+03	1.50e+03	2.23e-03	2.56e-02	5.87e-03	7.47e-03	4.93e-03	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F22022	2022	10	2.54e+01	8.96e+01	5.12e+01	5.17e+01	1.22e+01	4.00e+00	8.92e+00	4.02e+00	4.60e+00	1.59e+00	3.99e+00	8.92e+00	3.99e+00	4.57e+00	1.60e+00
F32022	2022	10	1.38e-02	5.66e-02	3.42e-02	3.49e-02	1.01e-02	4.20e-10	5.72e-09	1.31e-09	1.69e-09	1.11e-09	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F42022	2022	10	6.30e+01	1.39e+02	1.04e+02	1.04e+02	1.83e+01	3.59e+01	7.83e+01	5.46e+01	5.56e+01	9.24e+00	1.40e+01	5.00e+01	3.00e+01	2.92e+01	8.63e+00
F52022	2022	10	1.03e+00	3.42e+00	2.23e+00	2.20e+00	4.96e-01	2.24e-06	1.23e-04	1.43e-05	2.16e-05	2.23e-05	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F62022	2022	10	3.29e+04	3.52e+05	9.68e+04	1.07e+05	5.38e+04	6.48e+00	2.57e+01	1.57e+01	1.62e+01	3.93e+00	7.60e-04	2.42e-01	4.42e-02	5.54e-02	4.48e-02
F72022	2022	10	6.05e+01	1.89e+02	1.05e+02	1.09e+02	2.71e+01	5.50e+00	3.85e+01	3.07e+01	3.03e+01	5.00e+00	4.52e-02	4.99e+00	1.86e+00	1.85e+00	1.16e+00
F82022	2022	10	3.99e+01	8.84e+01	5.28e+01	5.56e+01	1.08e+01	1.33e+01	2.71e+01	2.35e+01	2.28e+01	3.08e+00	9.77e-02	6.13e+00	9.50e-01	1.35e+00	1.22e+00
F92022	2022	10	9.78e+01	3.92e+02	3.76e+02	3.55e+02	5.51e+01	9.52e-03	3.59e+02	3.59e+02	2.32e+02	1.73e+02	0.00e+00	3.59e+02	3.59e+02	2.32e+02	1.73e+02
F102022	2022	10	2.37e+02	2.64e+02	2.45e+02	2.46e+02	6.10e+00	2.21e+02	2.21e+02	2.21e+02	2.21e+02	3.53e-02	2.21e+02	2.21e+02	2.21e+02	2.21e+02	1.11e-13
F112022	2022	10	2.01e+01	3.76e+01	3.02e+01	3.02e+01	3.91e+00	1.80e-03	8.63e-03	3.92e-03	4.14e-03	1.28e-03	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
F122022	2022	10	1.67e+02	1.71e+02	1.69e+02	1.69e+02	8.85e-01	1.64e+02	1.65e+02	1.64e+02	1.64e+02	5.04e-01	1.64e+02	1.65e+02	1.64e+02	1.64e+02	5.04e-01

DE — estadisticas por funcion, D = 50

error al optimo sobre 51 corridas · b=best w=worst med=median avg=mean sd=std

func	suite	dim	5k					50k					500k				
			b	w	med	avg	sd	b	w	med	avg	sd	b	w	med	avg	sd
F12014	2014	50	1.35e+09	2.82e+09	2.20e+09	2.15e+09	3.06e+08	3.58e+08	1.06e+09	7.26e+08	7.54e+08	1.42e+08	9.07e+07	2.30e+08	1.33e+08	1.41e+08	3.42e+07
F22014	2014	50	9.98e+10	1.65e+11	1.33e+11	1.36e+11	1.57e+10	2.06e+10	3.95e+10	2.91e+10	2.94e+10	5.22e+09	2.05e+06	3.73e+07	6.07e+06	8.31e+06	6.43e+06
F32014	2014	50	2.96e+05	5.08e+05	4.00e+05	3.93e+05	4.30e+04	8.90e+04	1.63e+05	1.20e+05	1.21e+05	1.60e+04	8.31e+02	4.26e+03	1.82e+03	1.95e+03	6.44e+02
F42014	2014	50	2.12e+04	4.58e+04	3.11e+04	3.18e+04	6.76e+03	2.05e+03	7.04e+03	3.54e+03	3.63e+03	8.37e+02	1.01e+02	1.16e+02	1.06e+02	1.07e+02	3.41e+00
F52014	2014	50	2.11e+01	2.13e+01	2.13e+01	2.13e+01	4.32e-02	2.11e+01	2.13e+01	2.12e+01	2.12e+01	3.35e-02	2.11e+01	2.12e+01	2.11e+01	2.11e+01	3.47e-02
F62014	2014	50	7.37e+01	7.96e+01	7.72e+01	7.70e+01	1.44e+00	5.14e+01	7.28e+01	5.93e+01	5.95e+01	4.98e+00	2.04e+01	5.45e+01	2.97e+01	3.03e+01	6.14e+00
F72014	2014	50	9.24e+02	1.74e+03	1.38e+03	1.36e+03	1.81e+02	1.78e+02	4.72e+02	3.05e+02	3.09e+02	6.31e+01	8.70e-01	1.13e+00	1.04e+00	1.04e+00	4.42e-02
F82014	2014	50	6.21e+02	8.14e+02	7.10e+02	7.06e+02	3.88e+01	4.59e+02	5.57e+02	5.10e+02	5.10e+02	1.86e+01	1.99e+02	3.95e+02	2.98e+02	2.91e+02	4.46e+01
F92014	2014	50	7.62e+02	1.01e+03	8.82e+02	8.80e+02	5.70e+01	5.15e+02	6.47e+02	6.00e+02	5.93e+02	2.97e+01	3.88e+02	4.71e+02	4.38e+02	4.37e+02	2.01e+01
F102014	2014	50	1.24e+04	1.51e+04	1.42e+04	1.41e+04	5.43e+02	1.16e+04	1.31e+04	1.25e+04	1.25e+04	3.46e+02	4.72e+03	1.02e+04	7.60e+03	7.67e+03	1.21e+03
F112014	2014	50	1.31e+04	1.58e+04	1.51e+04	1.49e+04	5.54e+02	1.31e+04	1.47e+04	1.40e+04	1.40e+04	3.66e+02	9.58e+03	1.37e+04	1.31e+04	1.28e+04	8.65e+02
F122014	2014	50	3.85e+00	5.66e+00	4.70e+00	4.71e+00	4.12e-01	3.13e+00	4.56e+00	3.87e+00	3.92e+00	3.49e-01	1.80e+00	3.90e+00	3.40e+00	3.32e+00	4.00e-01
F132014	2014	50	6.11e+00	8.99e+00	7.98e+00	7.86e+00	6.04e-01	2.16e+00	4.16e+00	3.54e+00	3.43e+00	4.43e-01	6.04e-01	9.24e-01	7.76e-01	7.81e-01	6.70e-02
F142014	2014	50	2.39e+02	5.08e+02	3.98e+02	3.84e+02	5.12e+01	5.45e+01	1.25e+02	8.70e+01	8.87e+01	1.49e+01	2.73e-01	1.30e+00	4.75e-01	6.43e-01	3.32e-01
F152014	2014	50	8.97e+06	6.84e+07	2.65e+07	2.93e+07	1.13e+07	4.50e+04	1.50e+06	3.60e+05	4.32e+05	2.93e+05	3.18e+01	4.43e+01	3.83e+01	3.86e+01	2.31e+00
F162014	2014	50	2.32e+01	2.39e+01	2.36e+01	2.36e+01	1.76e-01	2.26e+01	2.35e+01	2.32e+01	2.32e+01	2.13e-01	2.21e+01	2.31e+01	2.28e+01	2.27e+01	2.07e-01
F172014	2014	50	6.19e+07	2.49e+08	1.60e+08	1.60e+08	4.20e+07	2.41e+07	8.81e+07	4.98e+07	5.09e+07	1.39e+07	1.03e+07	3.35e+07	1.78e+07	1.79e+07	4.50e+06
F182014	2014	50	2.30e+09	6.61e+09	4.74e+09	4.58e+09	9.86e+08	3.95e+08	2.34e+09	1.23e+09	1.28e+09	4.77e+08	5.23e+07	6.70e+08	3.82e+08	3.58e+08	1.72e+08
F192014	2014	50	2.81e+08	2.32e+09	1.22e+09	1.17e+09	5.01e+08	1.68e+06	2.12e+08	6.77e+07	7.44e+07	5.13e+07	5.87e+04	2.52e+07	3.87e+06	6.25e+06	6.56e+06
F202014	2014	50	6.41e+12	4.14e+14	8.19e+13	1.04e+14	8.69e+13	9.54e+09	3.79e+12	4.35e+11	6.59e+11	7.41e+11	1.40e+07	5.43e+10	6.77e+09	1.10e+10	1.24e+10
F212014	2014	50	4.42e+08	1.36e+09	8.31e+08	8.65e+08	2.32e+08	4.95e+07	2.91e+08	1.33e+08	1.37e+08	5.35e+07	8.00e+06	3.79e+07	2.00e+07	2.10e+07	5.82e+06
F222014	2014	50	8.58e+10	1.09e+13	2.01e+12	3.24e+12	2.96e+12	2.95e+08	1.13e+11	1.15e+10	1.73e+10	2.06e+10	8.91e+05	9.18e+08	7.00e+07	1.26e+08	1.56e+08
F232014	2014	50	6.37e+02	1.27e+03	1.03e+03	1.01e+03	1.28e+02	3.21e+02	3.93e+02	3.56e+02	3.56e+02	1.51e+01	2.35e+02	2.36e+02	2.36e+02	2.36e+02	7.10e-02
F242014	2014	50	6.96e+02	9.16e+02	8.34e+02	8.29e+02	4.50e+01	4.44e+02	5.51e+02	4.95e+02	4.97e+02	2.43e+01	2.69e+02	3.02e+02	2.88e+02	2.87e+02	7.41e+00
F252014	2014	50	3.55e+02	4.59e+02	4.17e+02	4.10e+02	2.66e+01	2.58e+02	3.15e+02	2.92e+02	2.90e+02	1.28e+01	2.15e+02	2.50e+02	2.33e+02	2.33e+02	8.26e+00
F262014	2014	50	1.06e+02	1.08e+02	1.08e+02	1.07e+02	5.80e-01	1.02e+02	1.04e+02	1.03e+02	1.03e+02	3.94e-01	1.01e+02	1.01e+02	1.01e+02	1.01e+02	8.31e-02
F272014	2014	50	2.19e+03	2.42e+03	2.33e+03	2.32e+03	5.75e+01	1.77e+03	2.23e+03	2.05e+03	2.04e+03	9.35e+01	8.29e+02	1.74e+03	1.17e+03	1.19e+03	2.17e+02
F282014	2014	50	2.71e+03	4.95e+03	3.85e+03	3.83e+03	6.02e+02	1.57e+03	2.06e+03	1.77e+03	1.76e+03	1.02e+02	1.10e+03	1.44e+03	1.21e+03	1.23e+03	8.18e+01
F292014	2014	50	6.97e+08	2.97e+09	1.94e+09	1.98e+09	5.64e+08	1.13e+08	5.81e+08	2.46e+08	2.69e+08	1.09e+08	5.00e+05	1.49e+08	1.21e+07	2.54e+07	3.27e+07
F302014	2014	50	6.89e+09	1.50e+12	2.19e+11	3.66e+11	3.54e+11	8.64e+08	2.05e+10	4.56e+09	5.69e+09	3.69e+09	5.76e+07	1.64e+09	5.34e+08	5.95e+08	3.28e+08
F12017	2017	50	8.79e+10	1.66e+11	1.39e+11	1.36e+11	1.66e+10	2.02e+10	4.90e+10	3.12e+10	3.24e+10	6.38e+09	2.29e+06	3.21e+07	7.94e+06	9.43e+06	5.88e+06
F22017	2017	50	2.00e+05	3.62e+05	2.94e+05	2.91e+05	3.05e+04	1.02e+05	1.81e+05	1.40e+05	1.38e+05	1.68e+04	1.23e+03	8.68e+03	4.59e+03	4.66e+03	1.84e+03
F32017	2017	50	1.68e+04	4.24e+04	2.84e+04	2.95e+04	5.59e+03	1.97e+03	4.95e+03	3.38e+03	3.38e+03	7.92e+02	7.57e+01	1.80e+02	1.65e+02	1.55e+02	2.41e+01
F42017	2017	50	1.13e+05	1.93e+05	1.55e+05	1.53e+05	1.61e+04	2.01e+04	4.57e+04	3.31e+04	3.30e+04	6.16e+03	4.38e+02	5.29e+02	4.75e+02	4.78e+02	2.28e+01
F52017	2017	50	7.11e-02	1.21e-01	9.23e-02	9.48e-02	1.40e-02	3.82e-02	7.93e-02	5.94e-02	5.87e-02	9.74e-03	8.58e-03	4.17e-02	2.25e-02	2.37e-02	7.54e-03
F62017	2017	50	3.26e+06	6.20e+06	5.00e+06	4.97e+06	6.35e+05	5.78e+05	1.39e+06	1.02e+06	1.00e+06	1.79e+05	5.58e+02	1.05e+03	6.97e+02	7.25e+02	1.09e+02
F72017	2017	50	1.49e+03	1.88e+03	1.70e+03	1.69e+03	9.07e+01	9.92e+02	1.35e+03	1.18e+03	1.18e+03	6.50e+01	7.09e+02	1.00e+03	8.80e+02	8.84e+02	5.41e+01
F82017	2017	50	8.24e+01	1.66e+02	1.41e+02	1.38e+02	1.62e+01	3.70e+01	6.95e+01	5.26e+01	5.30e+01	8.37e+00	7.75e-02	1.33e+00	2.86e-01	3.46e-01	2.18e-01
F92017	2017	50	1.43e+04	1.59e+04	1.53e+04	1.52e+04	3.98e+02	1.31e+04	1.50e+04	1.44e+04	1.43e+04	4.45e+02	1.25e+04	1.43e+04	1.34e+04	1.34e+04	3.97e+02
F102017	2017	50	8.75e+08	5.90e+09	3.77e+09	3.72e+09	1.24e+09	2.73e+07	2.22e+08	8.02e+07	9.23e+07	4.57e+07	4.06e+04	2.08e+05	8.82e+04	1.00e+05	4.38e+04
F112017	2017	50	5.71e+09	1.50e+10	9.78e+09	1.03e+10	2.30e+09	8.52e+08	3.16e+09	1.50e+09	1.66e+09	5.23e+08	2.12e+07	1.32e+08	5.54e+07	6.15e+07	2.67e+07
F122017	2017	50	2.05e+10	5.90e+10	3.82e+10	3.96e+10	9.06e+09	1.58e+09	8.42e+09	3.84e+09	3.89e+09	1.34e+09	3.58e+05	1.48e+07	3.43e+06	4.08e+06	3.53e+06
F132017	2017	50	5.57e+06	4.90e+07	2.28e+07	2.40e+07	9.53e+06	6.53e+05	6.44e+06	2.43e+06	2.59e+06	9.99e+05	9.81e+03	2.81e+04	1.77e+04	1.78e+04	4.40e+03
F142017	2017	50	2.84e+09	1.38e+10	7.10e+09	7.11e+09	1.99e+09	7.14e+07	8.29e+08	3.64e+08	3.81e+08	1.68e+08	2.13e+05	5.45e+06	1.54e+06	1.73e+06	1.20e+06
F152017	2017	50	7.70e+07	2.85e+09	7.26e+08	8.27e+08	5.32e+08	1.64e+05	9.36e+05	3.71e+05	4.15e+05	1.65e+05	2.41e+01	5.21e+02	8.19e+01	1.49e+02	1.48e+02
F162017	2017	50	7.60e+08	1.10e+12	1.41e+11	2.16e+11	2.44e+11	1.56e+07	2.90e+09	2.27e+08	4.73e+08	5.75e+08	8.54e+03	2.92e+05	7.63e+04	7.88e+04	5.10e+04
F172017	2017	50	2.20e+07	1.28e+08	7.31e+07	7.33e+07	2.85e+07	6.74e+06	2.54e+07	1.36e+07	1.42e+07	3.91e+06	4.23e+05	1.70e+06	1.11e+06	1.11e+06	2.93e+05
F182017	2017	50	1.32e+11	2.59e+13	2.45e+12	4.21e+12	4.79e+12	2.82e+08	1.70e+09	8.00e+08	7.94e+08	2.94e+08	8.27e+02	7.21e+03	2.15e+03	2.48e+03	1.33e+03
F192017	2017	50	5.45e+03	1.10e+04	9.44e+03	9.37e+03	1.20e+03	2.62e+03	4.95e+03	3.70e+03	3.69e+03	4.52e+02	1.62e+02	1.24e+03	4.06e+02	4.96e+02	2.87e+02

func	suite	dim	5k					50k					500k				
			b	w	med	avg	sd	b	w	med	avg	sd	b	w	med	avg	sd
F202017	2017	50	8.57e+03	1.72e+05	1.30e+04	4.84e+04	5.64e+04	2.81e+03	3.91e+04	3.61e+03	1.02e+04	1.24e+04	1.55e+02	2.41e+03	6.93e+02	8.41e+02	6.50e+02
F212017	2017	50	1.18e+03	3.07e+03	1.87e+03	1.95e+03	4.52e+02	4.20e+02	5.82e+02	5.12e+02	5.15e+02	3.45e+01	1.97e+02	2.30e+02	2.08e+02	2.10e+02	6.29e+00
F222017	2017	50	5.16e+04	7.62e+04	6.37e+04	6.37e+04	6.52e+03	2.21e+04	2.72e+04	2.46e+04	2.46e+04	1.46e+03	3.71e+02	5.33e+03	4.99e+02	6.85e+02	8.95e+02
F232017	2017	50	2.72e+04	4.78e+04	3.74e+04	3.71e+04	4.60e+03	1.10e+04	1.57e+04	1.36e+04	1.35e+04	1.02e+03	3.16e+02	2.54e+03	3.90e+02	4.40e+02	3.05e+02
F242017	2017	50	9.55e+03	2.63e+04	1.79e+04	1.81e+04	4.01e+03	2.52e+03	4.34e+03	3.27e+03	3.33e+03	4.34e+02	7.24e+02	7.96e+02	7.43e+02	7.46e+02	1.42e+01
F252017	2017	50	4.87e+03	9.97e+03	6.52e+03	6.67e+03	1.15e+03	1.75e+03	1.93e+03	1.80e+03	1.82e+03	5.01e+01	1.20e+03	1.62e+03	1.58e+03	1.48e+03	1.62e+02
F262017	2017	50	1.15e+03	1.85e+03	1.42e+03	1.42e+03	1.41e+02	6.43e+02	8.49e+02	7.14e+02	7.19e+02	4.49e+01	4.91e+02	5.68e+02	5.10e+02	5.15e+02	1.67e+01
F272017	2017	50	2.16e+03	3.71e+03	2.98e+03	2.97e+03	4.11e+02	8.17e+02	1.13e+03	9.50e+02	9.49e+02	7.27e+01	5.34e+01	1.90e+02	8.93e+01	9.20e+01	2.63e+01
F282017	2017	50	9.22e+09	1.33e+12	1.37e+11	2.37e+11	2.64e+11	2.64e+08	2.63e+09	1.39e+09	1.44e+09	6.04e+08	5.24e+04	1.45e+06	2.98e+05	3.77e+05	2.50e+05
F292017	2017	50	1.03e+11	6.04e+12	8.11e+11	1.30e+12	1.32e+12	2.13e+09	1.03e+10	5.57e+09	5.41e+09	1.85e+09	7.38e+06	6.08e+07	1.96e+07	2.12e+07	1.21e+07
F12022	2022	20	2.11e+04	5.63e+04	3.99e+04	3.99e+04	7.66e+03	1.82e+03	7.63e+03	4.65e+03	4.89e+03	1.22e+03	1.39e-08	6.28e-07	9.45e-08	1.47e-07	1.40e-07
F22022	2022	20	4.81e+02	1.37e+03	8.82e+02	8.61e+02	2.22e+02	6.50e+01	1.22e+02	7.98e+01	8.31e+01	1.29e+01	4.49e+01	4.49e+01	4.49e+01	4.49e+01	1.63e-10
F32022	2022	20	3.03e-01	6.45e-01	4.78e-01	4.69e-01	6.72e-02	4.89e-03	2.02e-02	1.18e-02	1.18e-02	3.04e-03	0.00e+00	1.14e-13	0.00e+00	3.34e-14	5.23e-14
F42022	2022	20	2.45e+02	4.32e+02	3.80e+02	3.70e+02	3.76e+01	1.74e+02	3.01e+02	2.62e+02	2.61e+02	2.68e+01	6.00e+01	2.24e+02	1.97e+02	1.87e+02	3.55e+01
F52022	2022	20	8.46e+00	2.21e+01	1.52e+01	1.50e+01	2.88e+00	1.58e+00	4.39e+00	2.65e+00	2.78e+00	7.05e-01	3.83e-11	3.94e-09	3.09e-10	4.23e-10	5.71e-10
F62022	2022	20	1.51e+08	8.06e+08	3.78e+08	3.85e+08	1.69e+08	1.03e+07	4.86e+07	2.21e+07	2.48e+07	9.85e+06	7.82e+03	7.24e+04	2.92e+04	3.12e+04	1.65e+04
F72022	2022	20	5.66e+02	2.56e+03	1.27e+03	1.28e+03	3.53e+02	1.21e+02	4.95e+02	3.03e+02	3.02e+02	9.25e+01	2.80e+01	1.49e+02	7.09e+01	7.34e+01	2.37e+01
F82022	2022	20	4.52e+03	1.58e+07	5.93e+05	1.76e+06	3.14e+06	2.92e+02	2.63e+03	1.15e+03	1.20e+03	5.16e+02	2.76e+01	3.64e+01	3.26e+01	3.24e+01	1.88e+00
F92022	2022	20	5.07e+02	8.59e+02	6.47e+02	6.57e+02	8.88e+01	3.40e+02	3.57e+02	3.45e+02	3.45e+02	4.01e+00	3.36e+02	3.36e+02	3.36e+02	3.36e+02	6.34e-13
F102022	2022	20	5.34e+02	5.17e+03	1.53e+03	2.45e+03	1.83e+03	4.22e+02	4.22e+03	4.28e+02	6.09e+02	7.50e+02	3.95e+02	5.71e+02	3.95e+02	4.07e+02	4.15e+01
F112022	2022	20	1.30e+02	1.83e+03	2.48e+02	4.21e+02	4.53e+02	1.05e+01	2.21e+01	1.52e+01	1.53e+01	2.72e+00	1.11e-06	1.18e+01	2.99e-06	2.32e-01	1.65e+00
F122022	2022	20	2.69e+02	3.42e+02	3.00e+02	2.99e+02	1.58e+01	2.36e+02	2.70e+02	2.39e+02	2.42e+02	6.85e+00	2.35e+02	2.68e+02	2.38e+02	2.41e+02	6.75e+00